

2020 AMC 10A Problem 23 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 23. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 23

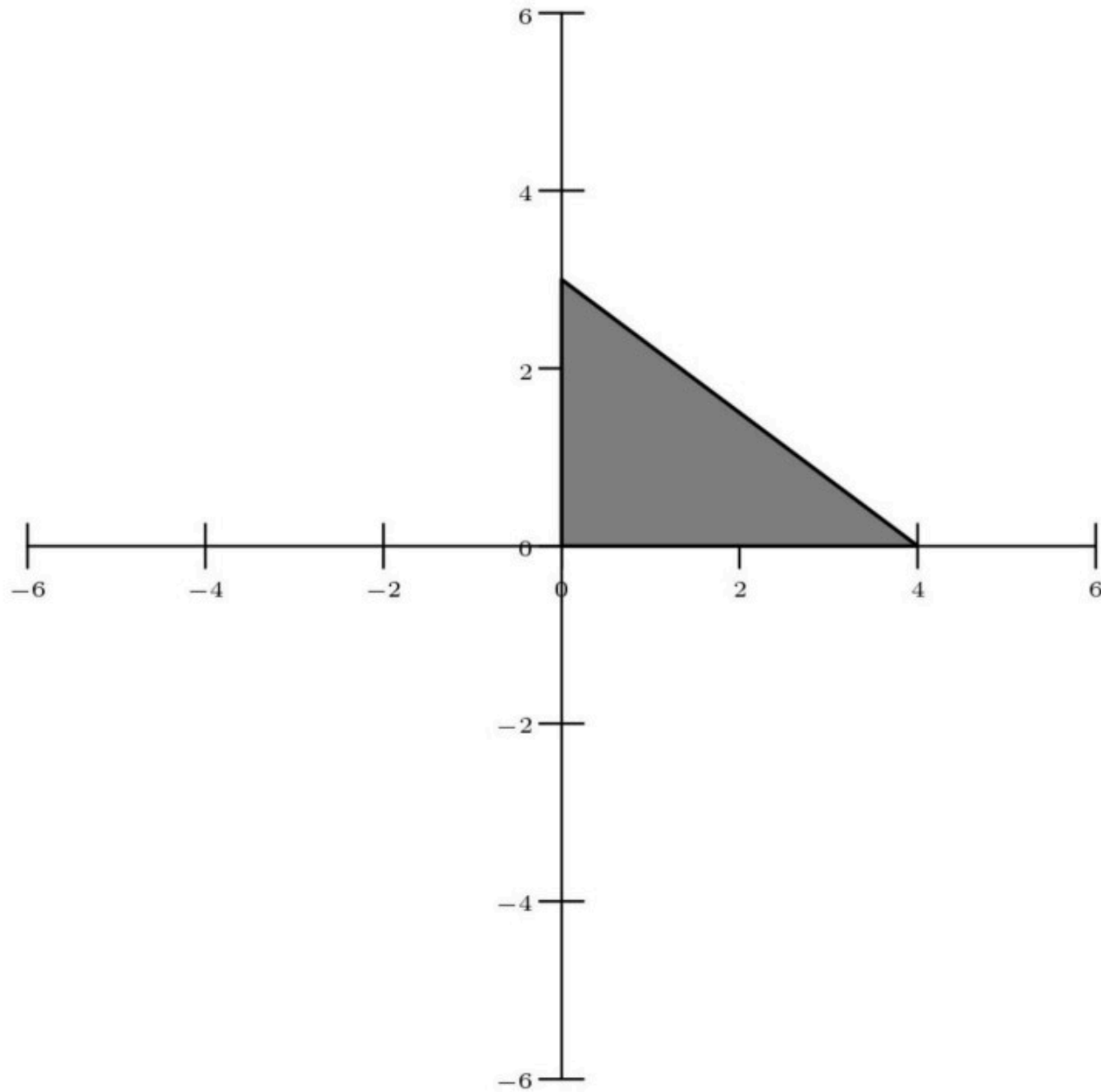
Problem:

Let T be the triangle in the coordinate plane with vertices $(0, 0)$, $(4, 0)$, and $(0, 3)$. Consider the following five isometries (rigid transformations) of the plane: rotations of 90° , 180° , and 270° counterclockwise around the origin, reflection across the x -axis, and reflection across the y -axis. How many of the 125 sequences of three of these transformations (not necessarily distinct) will return T to its original position? (For example, a 180° rotation, followed by a reflection across the x -axis, followed by a reflection across the y -axis will return T to its original position, but a 90° rotation, followed by a reflection across the x -axis, followed by another reflection across the x -axis will not return T to its original position.)

Answer Choices:

- A. 12
- B. 15
- C. 17
- D. 20
- E. 25

Solution:



First, any combination of motions we can make must reflect T an even number of times. This is because every time we reflect T , it changes orientation. Once T has been flipped once, no combination of rotations will put it back in place because it is the mirror image; however, flipping it again changes it

back to the original orientation. Since we are only allowed 3 transformations and an even number of them must be reflections, we either reflect T 0 times or 2 times.

► **Case 1:** 0 reflections on T .

In this case, we must use 3 rotations to return T to its original position. Notice that our set of rotations, $\{90^\circ, 180^\circ, 270^\circ\}$, contains every multiple of 90° except for 0° . We can start with any two rotations a, b in $\{90^\circ, 180^\circ, 270^\circ\}$ and there must be exactly one $c \equiv -a - b \pmod{360^\circ}$ such that we can use the three rotations (a, b, c) which ensures that $a + b + c \equiv 0^\circ \pmod{360^\circ}$. That way, the composition of rotations a, b, c yields a full rotation. For example, if $a = b = 90^\circ$, then $c \equiv -90^\circ - 90^\circ = -180^\circ \pmod{360^\circ}$, so $c = 180^\circ$ and the rotations $(90^\circ, 90^\circ, 180^\circ)$ yields a full rotation.

The only case in which this fails is when c would have to equal 0° . This happens when (a, b) is already a full rotation, namely, $(a, b) = (90^\circ, 270^\circ), (180^\circ, 180^\circ)$, or $(270^\circ, 90^\circ)$. However, we can simply subtract these three cases from the total. Selecting (a, b) from $\{90^\circ, 180^\circ, 270^\circ\}$ yields $3 \cdot 3 = 9$ choices, and with 3 that fail, we are left with 6 combinations for case 1.

- **Case 2:** 2 reflections on T . In this case, we first eliminate the possibility of having two of the same reflection. Since two reflections across the x -axis maps T back to itself, inserting a rotation before, between, or after these two reflections would change T 's final location, meaning that any combination involving two reflections across the x -axis would not map T back to itself. The same applies to two reflections across the y -axis. Therefore, we must use one reflection about the x -axis, one reflection about the y -axis, and one rotation. Since a reflection about the x -axis changes the sign of the y component, a reflection about the y -axis changes the sign of the x component, and a 180° rotation changes both signs, these three transformations composed (in any order) will suffice. It is therefore only a question of arranging the three, giving us $3! = 6$ combinations for case 2.

Combining both cases we get

$$6 + 6 = \boxed{(A) 12}$$

OR

Define s as a reflection, and r as a 90° counterclockwise rotation. Thus, $r^4 = s^2 = e$, and the five transformations can be represented as r, r^2, r^3, r^2s, s , and $rs = sr^{-1}$.

Now either s doesn't appear at all or appears twice. For the former case, it's easy to see that only r, r, r^2 and r^2, r^3, r^3 will work. Both can be permuted in 3 ways, giving 6 ways in total.

For the latter case, note that s can't appear twice, neither does r^2s , else we need to get e from r, r^2, r^3 , which is not possible. So r^2s and s must appear once each. The last transformation must be r^2 . A quick check shows that r^2, r^2s, s is permutable, since $r^2s = sr^{-2} = sr^2$ (since $r^4 = e$). This gives

6 ways.

Thus the answer is $\boxed{(A) 12}$.

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